AI Usage Journal

Principles of Computer Vision (ARI2129)

Group 4 - Watershed Segmentation

AI Usage Journal: Gradient Based Watershed

Student Name: Kurt Abela

Topic: Gradient-Based Watershed Segmentation

Date: May 1, 2026

Section 1: Tools Declared

- **Gemini 3 (Paid Tier)**
- **Google Forms:** Used to host the adversarial quiz.

Section 2: Notable Prompts

Prompt 1: The Deep Evolution Synthesis

> *"Explain both papers referenced and the topic gradient based watershed segmentation to me in a set of short notes and so I learn the material well"*

- **Summary of Response:** The AI synthesized the transition from the classical 1993 topography-based approach to the 2017 deep learning "Energy Map" approach, emphasizing why the latter handles non-convex shapes better.

Prompt 2: The Adversarial Construction

> *"Role: You are a high-level English Language speaker... Review my Q's and A's and flag where you think better English could have been used for more clarity and structure."*

- **Summary of Response:** The AI flagged all instances where my English was not up to standard or clear, it helped with this by providing alternative ways to structure my Questions and Answers.

Prompt 3: The "Perfect Score" Strategy

> *"Please explain the topic gradient watershed in a way I am consistently getting 10/10 in the pasted quiz."*

- **Summary of Response:** The AI provided a "topographical logic" guide, reframing mathematical formulas as physical metaphors (ridges, valleys, dams) to make the adversarial traps easier to spot.

Prompt 4: Feedback Loops

> *"Re-word and use better english for structure and clarity the following pasted answer feedback for each question."*

- **Summary of Response:** The AI again helped get the English up-to-standard and make the user feedback more understandable to users.

Prompt 5: Organisation

> *"Review the attached project guidelines and help organise our team of 5 to a set of tasks, each lending a hand in all tasks but taking lead in a single one"*

- **Summary of Response:** The AI helped distribute work equally and according to the marking scheme, which boosted our productivity.

Section 3: Five Questions

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

The distinction between the **Internal Gradient** and the **Beucher Gradient** was the most difficult. While the formula looks simple, explaining *why* the resulting watershed line shifts physically inside the object boundary requires a strong grasp of the material. Most people expect boundaries to sit "on" the edge, not inside it. It required multiple re-reads of the notes before understanding it completely.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

When helping with re-wording of the user feedback, the AI changed the context on one of the questions, which upon reading it I noticed this error and fixed it.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

The AI did not in any specific moment waste my time, as I used it for re-wording my English and explaining the topics based on the referenced pages, in which the AI responded accurately and fast.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

The AI helped greatly with organising our work and re-wording our work into professional English, also helped with the structure and neatness of all deliverables produced.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

Personally, the energy map regularisation, where some complex shapes are manipulated and if EDT treats them as multiple basins, the energy map treats them as a single parabolic basin. Although I understand the concept, when asked in detail about it, I get stuck.

AI Usage Journal: Gradient Based Watershed

Student Name: Tristin Bezzina

Topic: Alternatives & Code walk through

Date: 05/05/2026

Section 1: Tools Declared

- GPT 5.5 - Thinking

Section 2: Notable Prompts

Prompt 1: Roadmap to Understanding

“Which sections should I look at in the given paper so that I grasp the respective concept well of what it is trying to explain?”

- **Summary of Response:**

GPT 5.5 helped me identify the most relevant sections of the article instead of reading the whole paper blindly. It pointed me towards the sections that explained the main concept, the algorithmic idea, the problem being solved, and the relevance of the paper to watershed segmentation.

Prompt 2: Finding Alternative Segmentation Approaches

"Given the material of the watershed segmentation that i currently have what other alternatives can i go look for which applies segmentation in a different approach?"

- **Summary of Response:**

GPT 5.5 suggested that I look for segmentation techniques that are not just different versions of watershed, but methods that approach segmentation differently. It suggested alternatives such as graph cuts and active contours, where graph cuts frame segmentation as a global energy minimisation problem, while active contours use a curve that evolves towards the object boundary. I then used these suggestions to search for actual academic papers and checked whether the papers really supported the alternatives section before including them.

Prompt 3: Understanding how Distance Transform works

"Regarding distance transform, how does the method know which black pixel is the closest one to the current white pixel? Since the output distance transform keeps one value at the same position as the current foreground pixel, I want to understand how that value represents the distance between the current white pixel and its nearest black pixel”

- **Summary of Response:**

GPT 5.5 explained that distance transform conceptually finds the nearest black pixel like a nearest-neighbour problem, but uses an efficient implementation as it does not brute-force compare every white pixel with every black pixel but it computes the nearest-background distance through efficient image scans.

Prompt 4: OpenCV Methods for the Watershed Pipeline

"What OpenCV methods can I use to apply this watershed segmentation pipeline: original image -> grayscale -> thresholding and morphological cleanup -> distance transform -> foreground and background markers -> gradient image -> watershed segmentation?"

- **Summary of Response:**

GPT 5.5 suggested which OpenCV functions match each stage of the pipeline.

For grayscale conversion, it suggested cv2.cvtColor(). For thresholding, it suggested cv2.threshold() with Otsu’s method.

For morphological cleanup, it suggested cv2.morphologyEx() with opening and closing, together with a kernel created using np.ones() or cv2.getStructuringElement().

For distance transform, it suggested cv2.distanceTransform().

For creating foreground markers, it explained thresholding the distance transform and then using cv2.connectedComponents() to label marker blobs.

For background markers, it suggested dilation using cv2.dilate().

Prompt 5: Understanding the Watershed Flooding Process

"Explain the watershed flooding process in simple terms. I want to understand how labelled basins grow, how a pixel receives a basin label, and how the algorithm knows when two basins meet and should create a watershed boundary"

- **Summary of Response:**

GPT 5.5 explained that the flooding process uses a separate marker/label matrix, not the original pixel-value image. The image or gradient values guide the order in which pixels are processed, usually from low values to high values, while the label matrix stores which basin each pixel belongs to. If a pixel is reached by one labelled basin, it receives that basin's label. If it is reached by two different basins, it is marked as a watershed boundary. This helped me understand that watershed does not change the original pixel values; instead, it grows labels across the image and creates boundaries where different labels meet.

Section 3: Five Questions

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

The hardest concept to explain was how the landscape analogy of watershed segmentation maps to what is actually happening with pixel intensity values, gradient values, basins, labels, and boundaries.

At first, I understood the general idea that an image can be treated like a 3D landscape, where low areas are basins and high areas are ridges. However, the difficult part was understanding that the "height" is actually an intensity or gradient value, and that the flooding process does not change the original pixel values. Instead, those values guide the order of processing, while a separate marker/label matrix stores which basin each pixel belongs to.

I think this was difficult because the metaphor is easy to picture, but the actual implementation is more abstract. It involves matrices, labels, and neighbourhood checks rather than real water or real terrain.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

At one point, I asked GPT 5.5 for alternatives to watershed segmentation, but because I did not yet have proper knowledge of the topic, I naively accepted the first type of "alternatives" it gave me. The suggestions sounded correct at the time, but later, once I understood the actual watershed segmentation algorithm and the full practical pipeline, I realised that some of the suggestions were not true alternatives. They were mostly tweaks to the preprocessing pipeline that still ended up being passed into watershed.

This meant that I had given my colleague misleading information for the alternatives section. After realising this, I had to change my selected alternative papers and tell my colleague to fix that part of the notes again. I corrected the issue by asking more precise prompts, specifically asking for segmentation methods that are not watershed and that solve image segmentation using a different approach. This helped me identify more appropriate alternatives, such as graph cuts and active contours, and I then checked the actual papers before including them.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

A specific moment where GPT 5.5 wasted my time was when I asked for alternatives to watershed segmentation, but instead of giving me actual guidance on different segmentation techniques that could be used instead of watershed, it mainly suggested ways to tweak the preprocessing pipeline and still pass the result into watershed. For example, it focused on improving the watershed setup rather than clearly explaining alternative approaches such as methods that solve image segmentation in a different way. This made me realise that I needed to phrase my prompt more clearly by asking for "methods that are not watershed and apply segmentation using a different approach." I learned that AI can sometimes stay too close to the original topic unless the prompt clearly asks for a different direction.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

AI genuinely helped me work faster when I was trying to understand which parts of each article were relevant to our topic. Instead of reading every paper from start to finish without direction, I used AI to help identify which sections were likely to explain the main concept, method, limitations, or application. This made the research process faster because I could focus immediately on the most important parts of the paper. AI was also useful for summarising difficult concepts in simple terms, such as marker-controlled watershed, distance transform, over-segmentation. The difference was that AI helped me connect the theory to the practical notebook pipeline, so I could understand not only what the paper said, but also how it related to the actual implementation

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

One question we still do not feel completely confident answering is how to choose the best preprocessing parameters for watershed segmentation across every type of image. Our simulator helps us test parameters such as blur kernel size, thresholding method, morphology iterations, distance transform threshold, marker extraction, and noise filtering. This makes it easier to visually compare results and understand how each parameter affects the final segmentation. However, there is still no single set of values that works best for every image. The best parameters depend on the image type, noise level, object size, object shape, lighting, contrast, and whether objects are touching or separated. Therefore, even though the simulator helps us tune and understand the parameters, parameter selection still requires experimentation and image-specific judgement.

AI Usage Journal: Gradient Based Watershed

Student Name: Luke Camilleri

Topic: Topographic Metaphor, notes

Date: 05/05/2026

Section 1: Tools Declared

- Google scholar - Used to upload the key reference papers and query them interactively while writing
- Claude - Used to upload the key reference papers and query them interactively while writing
- NotebookLM - Used to upload the key reference papers and query them interactively while writing

Section 2: Notable Prompts

Prompt 1:

> *"Explain the topographic metaphor used in watershed segmentation. How does a greyscale image become a 3D landscape, and why this framing is useful for image segmentation rather than treating the image as a flat signal "*

- **Summary of Response:**

  The AI gave a clear, layered explanation: pixel intensity maps to elevation, bright regions become hills, dark regions become valleys, and boundaries between objects correspond to the ridgelines that divide drainage basins. It used the rainfall and flooding analogy to make the geometry intuitive before introducing any formal notation. This explanation became the backbone of Section 2.1 in the study notes.

Prompt 2:

> *"What is the formal mathematical definition of a regional minimum in the context of the watershed transform? How is it different from a local minimum, and why does the distinction matter? "*

- **Summary of Response:**

Claude explained that a regional minimum is a connected flat region M at height c such that every pixel adjacent to M but outside it is strictly higher, crucially a connected set, not a single pixel. It contrasted this with a local minimum which can be an isolated single pixel and may not be flat. This distinction matters because each regional minimum seeds exactly one catchment basin, so conflating the two concepts leads to incorrect predictions about how many basins the algorithm will produce. This shaped the Key Definitions subsection

Prompt 3:

> *""I need to write about Beucher's historical contribution to watershed segmentation for my study notes. Can you summarise what he established, what framework he worked within, and why it mattered for the field?"*

- **Summary of Response:**

Claude, gave a solid account of the CMM, the mathematical morphology framework from Matheron and Serra, and Beucher and Lantuéjoul's late-1970s work. However, it initially attributed the full gradient-based watershed formulation to Beucher alone, omitting Meyer's co-authorship of the canonical 1993 paper. After noticing this, the actual Beucher & Meyer (1993) paper was checked on Google Scholar to correct the attribution before including it in the notes.

Prompt 4:

> *"Help me structure my study notes on the topographic metaphor so that they flow logically from intuitive to formal. I need to cover: the image-as-landscape analogy, the three key definitions (regional minimum, catchment basin, watershed line), and the flooding simulation."*

- **Summary of Response:**

The AI suggested a four-part ordering: start with the intuitive image-to-landscape framing, move to the formal definitions, then introduce the flooding simulation as a constructive bridge between intuition and algorithm, and close with Beucher's historical context. This progression, analogy first, formalism second, was adopted almost exactly and made the notes read much more naturally than a purely definition-led structure would have.

Prompt 5:

> *"Explain to me in simple terms why the watershed line W(f) is defined as the complement of all catchment basins, rather than just being described as the ridge pixels directly "*

- **Summary of Response:**

Claude explained that defining it as a complement is actually the more precise approach, trying to identify "ridge pixels" directly would require setting arbitrary thresholds for what counts as steep enough, which introduces subjectivity. The complement definition falls naturally out of the flooding construction: once every basin is fully labelled, whatever pixel is left over is by definition a boundary point. This cleared up a confusion I had been sitting with for a while and made the formal definition in the notes make sense rather than just being something I was copying down.

Section 3: Five Questions

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

The formal definition of the watershed line as the complement $W(f) = D \setminus CB(m_i)$ was $\cup$ consistently the hardest to communicate. My friends, intuitively understand “ridgelines between valleys”, but struggle with the negative definition: a boundary described as everything that is not a catchment basin feels backwards. The challenge is that most people expect a boundary to be defined positively, as something you locate, rather than as whatever is left over after the basins are removed. Every attempt to explain it required going back to the flooding analogy first before the complement definition clicked.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

When writing the notes and I asked Claude to help with my writing to make it more formal, I noticed at many times the AI included additional information which I never entered and added imaginary articles. I noticed this by going through the text and identifying the errors.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

Mostly suggesting articles which were not relevant to the topic at hand. I wouldnt say it was a huge time waster, as I simply glimpsed through the article and I could see that there was no mention of topographic metaphor. I learnt that I always have to double check the information given to me by AI as its not always correct or adequate.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

Honestly, the most useful moment was the structure suggestion from Prompt 4. My first instinct was to open the notes with the formal definitions straight away, regional minimum, catchment basin, all the notation because that felt like the "proper" way to write study notes. Claude suggested flipping it. Lead with the analogy, get the mental picture in place, then introduce the maths once it already makes intuitive sense. When I showed the draft to the rest of the team they said it was way clearer than they expected for a topic this heavy on notation. That reordering is what made the difference and I genuinely would not have thought to do it that way on my own.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

The thing none of us feel confident on is actually proving why the watershed line is guaranteed to be a closed, connected set. The notes state it as fact and we understand the intuition, if you fill all the basins, whatever is left must form a connected boundary around them, but if someone asked us to construct the formal topological argument from scratch we would struggle.

AI Usage Journal: Gradient Based Watershed

Student Name: Jake Desira

Topic: Markers, Over-Segmentation & Simulator

Date: 10/05/2026

Section 1: Tools Declared

- **GPT 5.5 Thinkin**g: Used for learning, reasoning, checking my understanding of watershed segmentation, marker generation, over-segmentation, distance transforms, and parameter tuning.

- **Claude Opus 4.7**: Used for coding support when implementing and improving the interactive Streamlit watershed simulator.

Section 2: Notable Prompts

Prompt 1:

"Give me a detailed understanding of marker based watershed"

- **Summary of Response:** GPT 5.5 explained that marker-controlled watershed is used because normal watershed can easily over-segment an image when there are many small intensity variations or noisy local minima. Markers give the algorithm controlled starting points, so instead of flooding from every small minimum, it floods from selected regions that are likely to represent real objects. This helped me understand why markers are central to reducing over-segmentation.

Prompt 2:

"This seems like it is over-segmenting right?"

- **Summary of Response:** GPT 5.5 helped me interpret the simulator output and explained that over-segmentation happens when too many markers are created. In the context of the app, this could be caused by the peak neighbourhood being too small, the peak minimum height being too low, or the minimum marker area being too small. This helped me connect the visual result to the parameters controlling marker generation.

Prompt 3:

"What about for this photo? I know it is not perfect and is a bit under-segmented."

- **Summary of Response:** GPT 5.5 helped me reason through why different images need different parameter settings. For images such as pills, rice grains and cells, the best settings depend on object size, object spacing, contrast, overlap, and background texture. It explained that under-segmentation usually means there are too few useful markers, while over-segmentation usually means too many markers or noisy markers are being created.

Prompt 4:

"Give me base presets for the different type of images I have. I will edit them to make them better."

- **Summary of Response:** Opus 4.7 helped me create starting parameter presets for the different image types in the simulator, such as cells, pills/capsules and rice grains. The response gave me reasonable baseline values for thresholding, morphology, marker extraction, peak detection, and noise filtering. These presets were not treated as final perfect values, but as starting points that I could then test and manually adjust inside the interactive GUI.

Prompt 5:

"Help me make the gui more user-friendly."

- **Summary of Response:** Opus 4.7 suggested ways to make the GUI easier to understand for users who are not already familiar with watershed segmentation. It recommended reducing how overwhelming the sidebar feels by separating basic controls from advanced controls, adding clearer labels and explanations, and helping users understand what each stage of the pipeline is showing. This helped me improve the simulator from being just a technical tool into something more suitable for learning and experimentation.

Section 3: Five Questions

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

The hardest concept to explain was how marker generation controls over-segmentation. At first, it is easy to say that “more markers means more regions,” but explaining why this happens in the actual watershed process is harder. The markers act as the initial labelled basins, and the watershed algorithm grows these labels outward. Therefore, if the marker stage creates too many seeds inside one object, the algorithm may split one real object into several labelled regions. If it creates too few seeds, multiple touching objects may be merged into one region. This was difficult because the problem is not only about the watershed algorithm itself, but also about how preprocessing, distance transform, local maxima detection, and noise filtering all affect the number and quality of markers.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

AI sometimes gave parameter suggestions that sounded reasonable but did not work well when tested visually in the simulator. For example, it could suggest lowering a marker threshold or reducing the peak neighbourhood to detect more objects, but in practice this sometimes created too many small markers and caused over-segmentation. The explanation was logically correct in general, but incomplete because the actual result depends heavily on the image. I fixed this by testing the suggestions directly in the Streamlit simulator and comparing the intermediate stages, especially the distance transform, sure foreground, marker labels, and final watershed result. I learned not to trust parameter advice without checking the visual output.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

A specific moment where AI wasted my time was when I asked for help tuning a demo image and kept changing parameters based on suggestions without first clearly deciding whether the result was over-segmented or under-segmented. This made me adjust too many sliders at once, which made it harder to know which parameter actually caused the improvement or problem. I learned that for watershed tuning, it is better to change one parameter at a time and observe the effect on the markers and final boundaries. AI is useful for explaining what each parameter does, but it cannot replace careful visual testing.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

AI genuinely helped when building and improving the interactive watershed simulator. GPT 5.5 helped me understand the reasoning behind each stage, such as thresholding, morphology, distance transform, marker extraction, unknown region creation, and over-segmentation control. Claude Opus 4.7 then helped with the coding side, especially in structuring the Streamlit app. The difference was that AI helped connect the theory to the implementation. Instead of only having a static explanation of watershed, we ended up with an interactive tool where users can change parameters and immediately see how the markers and segmentation boundaries change.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

One question I still do not feel fully confident answering is how to automatically choose the best marker-generation parameters for any new image. The simulator makes it easier to test values such as peak neighbourhood size, foreground threshold ratio, peak minimum height, minimum marker area, morphology settings, and dilation iterations. However, there is still no single setting that works well for every image. A good parameter choice depends on object size, noise, contrast, whether the objects are touching, whether the objects are circular or irregular, and whether the background has texture. I understand how to tune the parameters manually, but I would still struggle to explain a fully automatic method that always selects the best markers without trial and error.

AI Usage Journal: Gradient Based Watershed

Student Name: Nathan Vella

Topic: Watershed Segmentation (Failure Modes) and Presentation

Date: May 16, 2026

Section 1: Tools Declared

- Google Scholar used to search and find relevant papers on the topic
- Perplexity used to surface the papers and summarise also try to find and search for other papaers and solutions
- Claude Design used to draft the presentation and build and refine some visuals

Section 2: Notable Prompts

Prompt 1:

> *"I m formulating a presentation on watershed segmentation which is organised into these 7 segementations. This will be used to teach other students what is the best tone and strategy to tackle it"*

- **Summary of Response:** it mainly focused on the strategy of building up the knowledge and providing a worked example which I continued to build on to help me formulate it to be best represntable for the others to understand. And pointers to avoid suchas not over covering everything and not focus on formulas but rather the understanding.

Prompt 2:

> *"I need to build a PowerPoint deck for a group presentation on watershed. Help me start to build on it and build a skeleton.*
>
> *The deck needs to follow the structure of our study notes, which are organised into seven sections: (1) Introduction to segmentation, (2) Topographical metaphor, (3) Gradient-based watershed (4) Markers and over-segmentation, (5) Failure modes, (6) Alternatives like graph cuts and active contours, (7) Real-world applications.*

For a worked example specifically, use the 1D signal [3, 1, 2, 8, 2, 1, 3] from our notes. Show the gradient as a table with columns for pixel. The final gradient should be [2, 2, 7, 6, 7, 2, 2]. Then on the flood slide, show the 7-step decision table with columns for step, water level h, pixel, decision, and current state across all 7 pixels. The result is three basins (B1, B3, B2) with watershed pixels at p3 and p5

For the morphological gradient slide, the formula is g(f) = (f⊕B) − (f⊖B), with dilation defined as the local maximum and erosion as the local minimum.

For the markers slide, distinguish internal markers (inside foreground objects) from external markers.

For failure modes, cover three: over-segmentation, boundary leakage, and noise sensitivity

Generate the deck slide by slide so I can review each one before moving on.

This will be the basis on what. I will build on.

"

- **Summary of Response:** A skeleton slide deck was built and diagrams were given which more was built on then I kept building and refining the message and the way to present it as here I already knew what I wanted to present

Prompt 3:

"check references match correctly to what they are intended from notes"

- **Summary of Response:** found some mismatches were references were made to a wrong segment or when I mistook one name for another and got mixed up

Prompt 4:

"from the notes is everything there needed and is there anything else which we should add to the presentation"

- **Summary of Response:** it found two gaps in the slides that werent covered and that were quite important to cover which included smoothing and distance transform and some nice to haves which were taken into consideration but not added as they added fluff

Section 3: Five Questions

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

How watershed worked and I landed on giving an example and how best to give the example for the people to understand how it works better as the slides and notes just explain what watershed is but to actually see it working is another thing and helps you visualise better.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

Although it was limited it sometimes go tconfused especially whne given papers to see about the topic and easily could mis reference and as I was finding papers and sources on which to base out assignment on I was focused on papers in failure cases perplexity return manny papers which could not be accessed or url that didn t exist those were nulled and not taken into consideration.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

Over relying on AI and it not correctly eplaining it which led to things not fully adding up at the beginning but as I sat down and read from the paper directly and watched a video it thought me much more as AI wa sgathering or getting confused and misinterpreting. While it always sounded veryconfident it didn texplain the algorithm and meaning exactly.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

When I went to star the presentation although the plan was made I found it hard to begin and how to design it and very indesicive on what I should do. With explaingin it in a lot of detail and building it with it and refining it helped design a skeleton and that gave me a very good starting point where it also helped me build the visuals in powerpoint to help people understand. I thelped me with the push forward and built the skeelton as I don t usually use powerpoint so was hesitant how t start although material was all gathered.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

How reconstruction works we can describe it but implementing it and going into more depth about it is much harder .